

EYE IN THE SKY 'Switch', a high-altitude spy drone, made by Indian start-up ideaForge, deployed with the Indian army in Ladakh

10% increase from last year's defence budget—going from Rs 4.78 lakh crore to Rs 5.25 lakh crore

LOOKING INWARDS

68% of capital outlay for defence set aside for Make in India

Impact: Local purchases have a force-multiplier effect on local industry, seeding ecosystems and generating jobs in the domestic market

25% of defence R&D budget earmarked for private industry, start-ups and academia

Impact: Overcomes hurdle of access to capital for defence start-ups and MSMEs that pursue technology development

Nodal body for setting up testing and certification of defence systems and platforms

Impact: Enables faster processing of defence innovations and assists private R&D set-ups in getting equipment qualified

senior ministry of defence (MoD) official. Presently, such a body does not exist.

The 'buy indigenous' move is part of the rebooted Aatmanirbhar in defence drive, launched in May 2020. A PMO-supervised effort is currently underway to fuel the two engines of an indigenous defence industry—defence public sector undertakings and private industry. While orders for big-ticket items like self-propelled artillery and missiles can be met by Tier 1 industries, the government needs to hand-hold start-ups in niche areas like drones and unmanned systems. This is where the budget has made a bold move—it has set aside 25 per cent of the Defence Research and Development Organisation's (DRDO) Rs 11,981 crore budget for private industry, start-ups and academia. "The R&D base of the country can be expanded with a synergistic approach by all stakeholders—academia, R&D organisations and industry. This is vital to develop advanced systems within the country and for exports," says G. Satheesh Reddy, chairman, DRDO. Welcoming the move, Baba Kalyani, managing director, Bharat Forge, says this will "pave the way for investments in frontier technologies and capability development".

The government also announced its intent to establish a nodal body to set up testing and certification of defence system and platforms, to enable private developers to test their defence innovations quickly. It has allocated Rs 1,365 crore for developing military hardware prototypes under the 'Make' category of MoD acquisitions.

The idea for a start-up fund for defence is not a new one. Three years ago, the DRDO set up a 'technology development fund' to promote self-reliance in defence. The DRDO has so far sanctioned funds worth Rs 164 crore to 2,611 companies, which has resulted in 150 technologies being indigenised—marine desalinators for use onboard life rafts stored on warships, high-performance propulsion systems for small satellites and drones for carrying stores in high-altitude areas. Other initiatives, like the Innovations for Defence Excellence (iDEX) launched in 2018, have not been as successful. The iDEX was meant to hand-hold MSME start-ups, R&D institutes and academia. Four years later, it has yet to successfully deploy any cutting-edge technologies and none of its products have entered field trials.



"THE R&D BASE OF THE COUNTRY CAN BE EXPANDED WITH A SYNERGISTIC APPROACH BY ALL STAKEHOLDERS—ACADEMIA, R&D ORGANISATIONS AND INDUSTRY. THIS IS VITAL TO DEVELOP ADVANCED SYSTEMS WITHIN THE COUNTRY AND FOR EXPORTS"

– G. SATHEESH REDDY
Chairman, DRDO

This explains why a DRDO-steered model appears to have found favour with the government. Top defence officials say the government favours a model based on the US defence department's Defence Advanced Research Projects Agency (DARPA). This model, if it works out, will see the DRDO focus on cutting-edge weaponry like aircraft engines and hypersonic weapons while leaving niche technologies to Indian industry. The DRDO has been pushing a Development cum Production Partner (DcPP), which will allow the private sector to co-develop missile systems and have manufacturing rights as well. The agency will transfer technology and patents to the private sector free of cost to boost indigenous defence manufacturing.

"We can't do everything," admits a senior DRDO official. "What the industry can do, the DRDO won't." The fifth-generation Advanced Medium Combat Aircraft, being designed by the DRDO, will be a special purpose vehicle and involve the DRDO, HAL and two private companies that will receive up to 75 per cent of funding from the government.

Things may be changing within the monolithic defence ministry, but it remains to be seen how quickly it happens to be effective. ■